Cova v9 Render Pipeline — From-Scratch Research and Commit

Scope: render pipeline only (upload → empty-room → design-direction → redesign → keep/compose → verify → refine). Everything else (auth, Stripe, product matching, marketing, YC) is explicitly out of scope.

Reader contract: this document is designed so Claude Code can begin Phase 0 immediately without additional research. Confidence levels are stated honestly; where validation is required before committing, the Phase 0 bench-test is the gate, not my prose.

Headline commit (one line): `SAM 3` furniture mask + 12–20px shadow-dilation → `fal-ai/bria/eraser` empty-shell → **`fal-ai/nano-banana-pro/edit`** redesign with shell + 1–3 exemplars + Cova archetype LoRA stack (when FLUX path is selected) → SAM 3 ghost/drift gate → keep-object composite (LAB-L + MIXED/NORMAL seamlessClone) → HPSv2.1/DINOv2 rank → optional `fal-ai/flux-pro/v1/fill` rescue. Primary path clears the $0.50/render and 60s p50 budget with ~3× cost headroom. **Confidence without bench validation: 70%. With a passing Phase 0 bench on the 20-room set: target 95%.**

Section 1 — Cova codebase audit (Phase 0)

The single largest risk of Rounds 1–11 was that every new primitive was tested in isolation against what the existing Cova codebase already assumed. Phase 0 is an audit, not a build, and it produces **a written Go/No-Go report** before any new service ships. The audit has six tracks, each with a concrete deliverable.

Custom LoRA audit. For every `.safetensors` file in the monorepo, in `modal/` bakes, and in Cova's fal/Replicate accounts: capture **base model (FLUX.1-dev vs FLUX.2-dev vs SDXL vs SD1.5), rank, alpha, trigger tokens, training dataset size and source, training steps, creation date, and author.** Any LoRA whose base is SD1.5 or SDXL is marked "deprecated — do not wire into v9" because the recommended pipeline is FLUX-family. Evaluate each surviving LoRA on five fixed exemplar prompts at weights 0.4/0.7/1.0 and score with HPSv2.1; any LoRA that does not beat the FLUX.1-dev baseline at some weight is dropped. **Deliverable:** **`docs/audit/loras.md`** **with a kept/dropped table and the full evaluation grid.** See also Section 5 — if the audit surfaces fewer than 3 archetype-aligned LoRAs of acceptable quality, Phase 4 must include custom LoRA training (~$150–$200 budget for nine archetypes on fal `flux-lora-fast-training` / `flux-2-trainer`, per the research).

Design Director audit. The existing two-call Sonnet 4.6 → Haiku 4.5 Director produces a 1000+ field DesignBrief compiled to a 50–80-word FLUX prompt, with 11 discrete `STYLE_DNA_BLOCKS` that explicitly *prevent* blending. The audit must answer three questions: **(1) Is the 6D fingerprint math continuous and bijective?** Verify that `(warmth, visual_complexity, organicness,`

`density, traditionalism, character)` → softmax top-3 archetype weights yields monotonic response to slider changes. (2) **Can STYLE_DNA_BLOCKS be safely removed?** The blocks prevent blending, but v9's archetype-LoRA-blend approach (Section 5) *requires* blending. Rewrite the blocks from "pick one" to "blend weights that sum to 1.0"; ship under a feature flag. (3) **Is the duplicated web-TS + Modal-Python Director converged?** The two implementations almost certainly have drifted; the audit must diff emitted prompts across 100 briefs and assert byte-equivalence or fold one implementation into the other. **Deliverable: `docs/audit/director.md` with the 6D sanity report, a STYLE_DNA_BLOCKS migration plan, and a TS↔Python parity report.**

Modal service audit. AnySplat 3D reconstruction and Marble API integration are noted as dead code in memory notes; **confirm with `git log --since=60.days` + runtime logs, then delete.** SparkJS viewer is kept only if the refinement screen (Section 8) uses it; otherwise delete. Keep only: Modal A10G image encoder service (required for SAM 2 hybrid and for any self-hosted LaMa tertiary), SAM embedding cache writer, and any existing quality-metric workers. **Deliverable: `docs/audit/modal.md` with a keep/delete table, plus removed-LOC and cold-start-latency deltas.**

Supabase RLS audit. Every query on `projects`, `renders`, `refinement_sessions`, and any new `sam_embeddings` table must be `auth.uid()`-scoped. Write a negative test that authenticates as user A and attempts to SELECT/UPDATE/DELETE every user B row; all must fail. Storage bucket policies must prevent signed-URL forgery across users. pgvector migration from IVFFlat to HNSW already happened; verify `ef_construction`, `m`, **and** `ef_search` parameters, confirm the embedding dimension matches the encoder in use (SigLIP-so400m-patch14-384 for InstantX IP-Adapter = 1152-dim; CLIP-L/14 = 768-dim), and benchmark top-10 recall against ground truth. **Deliverable: `docs/audit/supabase.md` — RLS negative tests added to CI, HNSW parameters locked.**

Test suite audit. The 715-test suite certifies current behavior; v9 introduces new categories. Coverage gaps to fill during Phase 10: mask morphology round-trips (dilation idempotency, alpha-feather pixel preservation), drift metric identity tests (SSIM(x,x)=1, DINOv2 cosine(x,x)≥0.9999, CIEDE2000 sanity on known pairs), YOLO-World duplicate detection, PaddleOCR watermark denylist hits, LoRA weight-vector normalization and clamping tests, forbidden-style-name regex tests for every canonical name (see Section 7), Zod schema rejection tests, snapshot tests for `buildPrompt(brief)`, and an E2E nightly run of 50 golden rooms × 10 archetypes feeding the Supabase aggregator. **Deliverable: `docs/audit/tests.md` listing the ~40 new test cases to add in Phase 10, with a coverage gap-fill matrix.**

pgvector audit. Confirm HNSW params; profile the refinement-screen "find-similar-region" query; verify the SigLIP/CLIP embedding pipeline runs on Modal A10G at acceptable latency (<200ms); confirm query-time filtering (user_id + archetype) is indexed and does not fall back to sequential scan. **Deliverable: `docs/audit/pgvector.md` with EXPLAIN ANALYZE output for top-5 queries.**

Exit criterion for Phase 0: all six deliverables landed and reviewed; any LoRA that fails evaluation is moved to `/archive/loras/`; dead Modal services deleted from `deploy.py`; Director parity proven; RLS negative tests green in CI. **Only then** does Phase 1 begin.

Section 2 — Competitor deep reverse-engineering

2.1 Collov AI — pipeline reconstruction with calibrated confidence

Collov operates on a nucleus of published research (D-Edit AAAI 2025, Meissonic/Monetico ICLR 2025, FlexControl ICML 2025, ViewControl IJCAI 2024) that shares authors Jinbin Bai, Zhen Dong, Aosong Feng, Xiao Zhang, and Kaicheng Zhou between the Collov research brand (`github.com/collovlabs`) and the MeissonFlow/academic collective. The org has **only four public repos, all stale since Aug–Nov 2024** — production code sits in a private org and materially diverges from the research artifacts.

D-Edit (arXiv 2403.04880). Latent-diffusion UNet with a **grouped cross-attention** primitive: the image is partitioned into K "items" by SAM/GroundingDINO masks, and each item's patches attend only to a short item-prompt (two special tokens). Editing partitions the mask into hard (frozen latents), active (denoised with the edited prompt), and soft (boundary blend) regions. This is the structural explanation for Collov's MLS-grade "does not modify permanent fixtures" claim — hard-mask compositing on walls/floor/windows is ~95% certain to be in production, though the *per-image UNet fine-tune* of the paper is almost certainly replaced by a forward-feeding adapter (IP-Adapter + MasaCtrl style) to hit their <15s p50.

Monetico/Meissonic. ~1B parameter masked-image-modeling transformer with VQ-VAE tokenizer and MaskGIT-style parallel decoding; 48 steps at 1024^2 in ~12–13s FP16 on A10G-class VRAM. Runs on A10G — ~95% confident. **Whether Collov's production backbone is Meissonic-finetuned or SDXL/FLUX + ControlNet is the critical open question**, and I mark my prior at SDXL/FLUX+ControlNet ≈ 60%, Meissonic ≈ 20%, hybrid ≈ 15% based on the under-15s p50 marketing, the industry default, and Monetico's low HF download count (research artifact, not production).

FlexControl (arXiv 2502.10451). Differentiable router over copied-UNet-blocks, with a computation-aware loss that shuts off blocks per step. Enables cheap multi-ControlNet at ~20–50% of full-stack FLOPs. Code release was "coming soon" in Feb 2025; **no evidence of open release as of April 2026.**

ViewControl (IJCAI 2024). LLM-Planner → pose estimation → novel-view synthesis pipeline using LangSAM (GroundingDINO + SAM). The research ancestor of Collov's 360° panorama product; production has almost certainly moved to newer NVS/MV-diffusion backbones.

Reconstructed end-to-end Collov pipeline (confidence in parentheses):

1. **Upload + preprocess** — EXIF rotation, resize to 1024/1536, optional SR via separate "Enhance Photo Quality" endpoint (95%).
2. **Perception & masking** — GroundingDINO text prompts → SAM/SAM2 via LangSAM, plus monocular depth (MiDaS/Depth-Anything) and layout estimator for a "protected structure" mask (95%; `collov.ai/API` explicitly exposes Object Masking API).
3. **Empty-room / declutter** — deletion mask (furniture union, dilated) → large-mask inpainter, most likely SDXL-Inpaint or FLUX-Fill tuned on interiors (~70%) over Meissonic-inpaint.
4. **Redesign** — SDXL/FLUX + depth-ControlNet + seg-ControlNet + IP-Adapter style ref at 1024^2 / 20–30 steps on AWS g5 A10G (~60%).
5. **Item-level preservation** — D-Edit-style grouped cross-attention with hard/active/soft masks; productionized with a forward-feeding adapter rather than per-image fine-tune (~85% on mechanism, ~20% on literal AAAI code).
6. **Post** — color correction, SR upscale to "2K", MLS disclaimer watermark, CDN via AWS CloudFront (serves `d17axom7zrjq3q.cloudfront.net`).

Pricing signal: ~$0.17/photo on annual plans, $7/room for 360° panorama; Enterprise offers API access + custom style/furniture training. Engineering team is small (~15 headcount, ~1 dedicated MLE per RocketReach) — Cova will not out-hire them but cannot out-distribute them either.

What Cova must match: (a) <30s perceived turnaround with a 60s p50 SLA ceiling; (b) verifiable structural preservation with a visible "structure lock" guarantee; (c) breadth of first-class verbs (empty-room, add-furniture, swap-material, twilight, style-swap).

What Cova exploits: (a) **Generic photoreal quality ceiling** — Collov targets $0.17/photo throughput, not hero quality, and reviewers report "complex scenes sometimes need a couple of passes"; Cova's $0.50/render budget buys Nano Banana Pro or FLUX.1-dev at higher-step counts with archetype-LoRA polish. (b) **Shallow curated style diversity** — Collov runs generic style presets; Cova's 9 canonical archetypes with 6D fingerprint and trained LoRAs is a real wedge. (c) **No meaningful 3D/digital-twin depth** — Inman rated the 360 product as "not a fully fleshed-out digital twin"; not Cova's target this phase but a future wedge.

2.2 Interior AI (Pieter Levels)

Current stack as of April 2026 (inferred from his unusually transparent WIP log at `wip.co/projects/interiorai`): **Nano Banana 2 as primary empty-room/edit model**, Flux Kontext fallback, Qwen Image Edit second fallback, legacy SD1.5 + depth-ControlNet + Segformer/UPerNet kept for "Interior design (with 3D Vision)" mode, World Labs API for 3D walkthroughs, Kling 1.6 for video. March 2026 WIP entry literally says: *"change empty space model from Flux Kontext to Nano Banana 2"*. The brief's cached tweet ID 1820572843324882991

is actually the Aug 2024 "Flux Schnell on Photo AI" tweet; the correct depth+Segformer reveal is `x.com/levelsio/status/1631924830063964160` (March 2023). **This is the single most important finding for Cova's architecture choice**: Levels, running the category-defining indie product, switched his primary redesign model to Nano Banana 2 in March 2026, which corroborates our Section 4 recommendation of the same family.

The "photos from angles perform worse" problem is structural, not fixable by prompt engineering: depth estimators and Segformer/UPerNet both train on ADE20K-style wide frontal real-estate photography. Oblique phone shots produce corrupted ControlNet maps. Cova must either (a) prompt users to reshoot frontally, (b) pre-process with geometric rectification, or (c) train/finetune on oblique views. Ignoring guarantees a long-tail complaint channel.

2.3 ReimagineHome, Virtual Staging AI, and the Decor8 tell

ReimagineHome (owned by Styldod) has pivoted to agentic conversational UX in 2025–2026 (0 credits for chat, 1 credit for render, 2 credits for render+product match). Pricing $19–119/mo; 26 product modes. Parent Styldod supplies human-graded ground truth, probable quality moat.

Virtual Staging AI ships the cleanest API in the space: one endpoint with `declutter_mode: off|on|declutter_only`, optional user-drawn mask URL, `base_variation_id` that anchors subsequent staging variations for multi-view consistency, webhooks, up to 20 variations per render, 10-second turnaround on Enterprise. Backend lives at `us-central1-furniture-ai.cloudfunctions.net` — Google Cloud Functions + likely SDXL/FLUX-family inpainter with SAM/GroundingDINO masking. **Cova should copy this API ontology verbatim** for the render pipeline's internal orchestration interface.

Decor8 is the stack-honest tell. Their own developer blog states they use "Stable Diffusion, ControlNet, Computer Vision, Image segmentation, and up-scaling". Their API exposes a preserved-objects whitelist per request (`wall, ceiling, door, window, floor, built-ins, kitchen island, fireplace, chandelier`) — Cova should copy this list into its preservation prompt vocabulary. **Pedra.ai's 4-verb ontology** (Enhance / Remove / Furnish / Renovate) covers 95% of demand and is the cleanest taxonomic template.

2.4 Open-source reference implementations worth porting

StableDesign (github.com/Lavreniuk/generative-interior-design, 128★) won 2nd place at AiCrowd Generative Interior Design 2024 with **SD1.5 + dual-ControlNet (seg + depth) + LoRA (rank 64), trained on ~300k Airbnb listing images captioned by LLaVA-1.5 and segmented by openmmlab/upernet-convnext-small.** Key trick: convert the seg mask to binary (window=1 else=0) and use as **inpainting mask to preserve windows**. This single technique addresses Cova's "window flip" failure mode at near-zero cost. **Port this to FLUX ControlNet** — no one has published the dual-CN + binary-window-inpaint recipe on FLUX yet; it is a concrete Cova quick win.

BertChristiaens/controlnet-seg-room is SD1.5 seg ControlNet trained on 130k LAION-5B interiors, 15 room types × 30 styles, ADE20K UPerNet segmentation. Use for reference, but its ADE20K reliance breaks on non-canonical objects — exactly Cova's failure class.

Kujiale/controlnet-layout is the closest-in-kind open model: SD1.5 ControlNet explicitly trained to auto-complete furnishing of a room given structural elements. SD1.5 ceiling caps quality; useful only as an auxiliary signal or as ground truth for Cova's own FLUX.2 LoRA training.

2.5 Collov vs Cova — the commit

Cova wins on **aesthetic ceiling** (NB-Pro reasoning + archetype LoRA stack), **structural preservation as a verifiable contract** (SAM 3 ghost gate + SSIM/DINOv2 quantitative thresholds surfaced in the UI rather than asserted in copy), and **curated style diversity** (9 canonical archetypes on a 6D fingerprint). Collov wins on **distribution** (20k+ realtor accounts), **feature breadth** (13+ modes), and **speed marketing** (<15s advertised). Cova's p50 target of 60s is 3–4× Collov's marketed number and is the honest constraint; if the bench shows NB-Pro p50 ⩾ 40s on fal queues, Phase 5 must downgrade primary to Nano Banana 2 (Flash) or route to FLUX.2 Pro Edit whose DiT path runs 8–15s.

Section 3 — Empty-room generation

3.1 Commercial APIs — head-to-head

Endpoint	$/image (1MP)	Prompt-free?	Preservation	Ghost risk	Note
`fal-ai/bria/eraser`	$0.04 flat	✅	**High**	**Low**	Licensed-data training, commercial-safe, **not prompt-conditioned** — cannot hallucinate new architecture
`fal-ai/flux-pro/v1/fill`	$0.05/MP	❌	Low on rect masks	Med–High	Cova already failed here; prompt drift + rectangle-mask proportion damage

Endpoint	$/image (1MP)	Prompt-free?	Preservation	Ghost risk	Note
FLUX.2 Pro / Flex / Max (edit mode)	$0.03–0.10/MP	❌	Unpredictable on unmasked	Medium	**No dedicated FLUX.2 Fill on fal as of April 2026** — BFL has not shipped a mask-conditioned v2 Fill
`fal-ai/ideogram/v3/edit`	~$0.06–0.08	❌	Low	Medium	Same generative drift class
Runway Gen-4	deprecated legacy inpaint	—	—	—	Runway pushed users to Aleph/Gen-4 Image refs
Virtual Staging AI declutter	~$0.93–4.17 (subscription math)	✅	High	Low	Enterprise-only API; awkward per-call billing
Clipdrop/Cleanup (Jasper)	~$0.003/call tier	✅	Medium (LaMa-class)	Low-Med	Weaker on very large masks
Adobe Firefly	Opaque enterprise	optional	Medium-high	Low	Licensing friction

Decision: Bria Eraser is the primary empty-room endpoint. It is prompt-free, commercial-safe, flat-priced, and structurally cannot hallucinate new architecture — which directly addresses Cova's Round-2.5 failure mode ("every model drifted architecture… cannot be prompted out of training-distribution biases"). Because no prompt is accepted, there is no prompt-to-pixel drift surface. **Confidence: high on pricing/endpoint, medium-high on quality; the Phase 0 bench confirms.**

3.2 Research models as rescues

LaMa (advimman/lama) remains the de-facto open baseline: ~51M params, FP16 on A10G runs 1–3s at 1024px, Apache-2.0. Weak on very large masks that span full wall corners because it has no semantic knowledge. **Deploy self-hosted on Modal A10G as the tertiary fallback.**

Matterport "Defurnishing with X-Ray Vision" (arXiv 2506.05338, CVPR-W 2025) is the definitive structure-guided recipe: build a Simplified Defurnished Mesh, extract

depth+normal+Canny, feed as ControlNet conditioning to SD-Inpaint. Mesh is not available in Cova's single-photo flow, but the adaptation is immediate: **Depth-Anything-V2 → Canny-from-depth-discontinuities → `fal-ai/flux-general/inpainting` with Canny control weight ~0.7 and prompt "empty room, original architecture preserved".** This is the Section 3.5 secondary fallback path.

ObjectClear (arXiv 2505.22636, code released July 2025) is the best open model for removing objects *with their shadows and reflections* — outperforms PowerPaint, RORem, OmniEraser on its benchmark. **NTU S-Lab non-commercial license blocks production use.** If licensing is relaxed in Q2/Q3 2026, displace Bria. Meanwhile the shadow-leak problem is mitigated by Cova's **12–20px morphological mask dilation before erasure** (Section 3.3).

Geomagical "Layout Aware Inpainting" (arXiv 2210.15796) introduced per-plane rectified inpainting — the exact antidote to Cova's "rectangle masks destroy proportions" failure. The public `github.com/geomagical/lama-with-refiner` refinement branch is a practical base for the tertiary path.

Matterport "An Empty Room Is All We Want" (arXiv 2405.03682) and Geomagical Layout-Aware are design patterns, not shipped weights.

3.3 Perception for furniture masking

SAM 3 (`facebook/sam3`, Nov 19 2025) is the single largest post-Round-9 enabler. Promptable Concept Segmentation with open-vocabulary short noun phrases (SA-Co benchmark, 270k concepts, 75–80% of human performance). **Available on fal at `fal-ai/sam-3/image` at $0.005/request, marked commercial-use.** SAM 3.1 (`fal-ai/sam-3-1/image`) is ~7× faster on multi-object workloads. **This directly solves Round 9's failure** ("SAM failed on synthetic empty canvases, coverage 9–18% vs 25–40%") because text prompts eliminate the need for seed points on synthetic canvases: `prompt="sofa"` returns zero masks on a blank canvas and tight silhouettes on a furnished one. **Confidence: high on capability and fal availability; medium on A10G self-host latency (not published — plan ~0.5–2s at 1024px).**

SAM 2 is retained only for the browser-hybrid tap-to-keep path (Modal A10G encodes once, caches embeddings, the ONNX-web decoder runs client-side <50ms per tap; Labelbox-style pattern). **GroundingDINO + SAM** is the pre-SAM-3 pattern; keep as self-hosted fallback if SAM 3 licensing on weights becomes a production blocker. **YOLOE** (`THU-MIG/yoloe`, ICCV 2025) at ~20–40ms/image on A10G is the cheapest open-vocab detector; use as a pre-filter or latency-critical fallback. **Segformer/UPerNet on ADE20K** is the reliable closed-vocabulary baseline for "everything that isn't wall/floor/ceiling/window/door" derivation and runs in ~100ms on A10G.

3.4 Empirical validation — the 20-room bench protocol

Phase 0 picks **8 of the 20 rooms stratified across four dimensions**: clutter density (2 light, 2 medium, 2 heavy), geometry difficulty (≥1 long perspective, ≥1 corner, ≥1 ceiling), texture

regularity (≥1 patterned floor, ≥1 textured wall, ≥1 plain wall), lighting (≥1 harsh directional, ≥1 soft overcast, ≥1 night/lamp-lit).

Metrics and acceptance thresholds per image:

- **SSIM-unmasked** on non-furniture architecture pixels: ≥ **0.985** (anything lower signals drift into preserved regions).
- **Ghost-furniture detection** by second-pass SAM 3 on the output with `["sofa","chair","table","bed","lamp","rug","cabinet","plant"]`: any returned mask with area > **0.5%** of the image fails.
- **Perspective consistency**: vanishing-point shift (LSD line detector) < **2%** of image width.
- **Edge-density ratio** at mask boundary (Canny inside vs outside a 16px ring): **0.7–1.3** vs clean reference.
- **Latency**: p50 ⩽ 60s, p95 ⩽ 90s.
- **Cost**: ⩽ $0.50/render.

Rollup accept: ≥7/8 rooms pass AND mean SSIM-unmasked ≥ 0.99.

3.5 Final empty-room recommendation

```
Stage E1  SAM 3 text-prompted furniture mask  ($0.01–0.03)
Stage E2  Morphological dilation +12–20 px (captures shadows/reflections)
Stage E3  fal-ai/bria/eraser  ($0.04)                  ← PRIMARY
Stage E4  Ghost gate: SAM 3 rerun + SSIM + VP-shift
Stage E5  If fail: Depth-Anything-V2 → Canny-from-depth →
          fal-ai/flux-general/inpainting w/ Canny CN @0.7  ← SECONDARY
Stage E6  If still fail: self-hosted LaMa on Modal A10G    ← TERTIARY
```

Total cost primary path ≈ $0.05/render. Worst-case triple-fallback ≈ $0.15. Well inside $0.50. Primary latency p50 ≈ 20–35s.

Section 4 — Redesign on the empty canvas

4.1 Head-to-head instruction-edit comparison

Endpoint	$/1K edit	Max res	Multi-ref	Reasoning	Arch-preservation	Es
`fal-ai/nano-banana-pro/edit` (Gemini 3 Pro)	$0.15	4K ($0.30)	14	Gemini 3 Pro — **strongest**	Native semantic planning	15 (u
`fal-ai/nano-banana-2/edit` (Gemini 3.1 Flash)	$0.08	4K ($0.16)	14	Gemini 3.1 Flash	Semantic, faster	5–
`fal-ai/flux-2-pro/edit`	$0.03 (1 MP)	4 MP	9 + `@` syntax	Mistral-3 VLM	Pixel-precise, geom lock + HEX color	8–
`fal-ai/flux-2-max/edit`	$0.07/MP	4 MP	9 + `@`	Mistral-3 VLM	Highest quality	15
`fal-ai/flux-pro/kontext/max`	$0.08	~2 MP	1 (/multi)	DiT	**Weak (Cova confirmed)**	9–
`fal-ai/bytedance/seedream/v4.5/edit`	$0.04	4 MP	10	Unified gen+edit	**Loose (Cova confirmed)**	2–
`fal-ai/qwen-image-2/pro/edit`	$0.075	~1 MP	Multi	Qwen2.5-VL + VAE	Dual-encoder preservation	~1

Primary redesign recommendation: `fal-ai/nano-banana-pro/edit` with shell-image + 1–3 exemplar multi-ref. Rationale mapped directly to each of the 11 failure modes:

Failure	Why it happened	Why NB-Pro fixes it
R1/R2 window flips / floor swaps / wall drift	Kontext/Seedream treat preservation as a soft prompt token	NB-Pro's Gemini 3 Pro reasoning plans architectural preservation as a precondition before rendering a pixel
R3 richer prompts break geometry	Prompt tokens compete with geometry signal	Exemplar image carries aesthetic density; prompt stays sparse + preservation-focused
R4 depth CN hallucinated windows	Depth is flat on walls	Depth removed from primary path; Canny/MLSD only in secondary
R4 flux-general 1-ControlNet limit	Platform constraint	NB-Pro has no ControlNet dependency; FLUX-general rescue uses Union-Pro-2.0 (multi-mode internally) + IP-Adapter slot + fill slot (orthogonal)
R5 empty-canvas bizarre content	Unconditioned generation	All recommended paths condition on the shell image; never a blank start
R7/R8 paste-back ghost furniture	Paste-back from input	Empty-canvas source from Bria eraser + SAM 3 ghost-gate rejects leakage
R9 SAM failed on synthetic canvases	Point/box seeds need salience	SAM 3 text-prompted segmentation works on any canvas
R10 double-Fill decor too sparse	Tight furniture masks leave no floor-free decor space	Instruction-edit generates dense decor over the full interior volume of the empty shell
R11 user's "clear then redesign"	Correct intuition	This IS the recommended pipeline

Confidence on NB-Pro primary: 75% pre-bench, 95% if bench passes. The open empirical question is NB-Pro's p50 on fal queues — fal explicitly does not promise speed for Pro-tier models. If bench p95 > 60s, demote NB-Pro to quality-tier (paid premium) and run Nano Banana 2 Flash as default primary.

Fallback: `fal-ai/flux-2-pro/edit` with shell + exemplar via `@image1`/`@image2` syntax, plus HEX color lock extracted from exemplar LAB-dominant clusters. FLUX.2 Pro Edit costs ~$0.03–0.06/render at 1–2 MP and runs typically faster than NB-Pro. HEX color control is unique to FLUX.2 and directly forces palette coherence — a Cova v9 differentiator.

4.2 ControlNet-guided fresh generation (rescue only)

`fal-ai/flux-general` supports **exactly one ControlNet at a time** (confirmed April 2026) — this is the Round 4 platform constraint. **But: Shakker-Labs FLUX.1-dev-ControlNet-Union-Pro-2.0** internally fuses modes (canny, soft-edge, depth, pose, gray) in *one* ControlNet weight; combined with the orthogonal `ip_adapter`, `easycontrol`, and `fill_image` slots, this yields effective multi-condition control without violating the platform rule. Recommended conditioning scales: canny 0.7 end 0.8; depth 0.8 end 0.8. **Depth-Anything-V2-Metric-Indoor-Large-hf (335M, DPT + DINOv2, Hypersim fine-tune)** is the correct indoor depth model and runs 1.5–3s on A10G at 1024px.

4.3 Specialized and reference models

RoomDiffusion (arXiv 2409.03198) is paper-only — no public checkpoint. Cite as aspirational benchmark; its multi-aspect training + LCD techniques apply if Cova trains its own LoRA. **Monetico/Meissonic** is too low-res (512^2) for production. **Kujiale/controlnet-layout** is the closest-in-kind open model but stuck on SD1.5.

IP-Adapter on FLUX: InstantX/FLUX.1-dev-IP-Adapter exists (SigLIP-so400m encoder, 128 tokens, injected into 38+19 blocks) but the InstantX page itself warns it is "not ideal for fine-grained style transfer." **XLabs-AI/flux-ip-adapter-v2** is generally preferred by community for style transfer. **`fal-ai/flux-general` exposes one IP-Adapter slot** with `image_url` + `scale` + optional `mask_url`; multiple IP-Adapters stack. For true multi-image reference, use **FLUX.2's native `@image1..@image10` syntax** or NB-Pro's 14-reference array.

FLUX.1 Redux (`fal-ai/flux-pro/v1.1/redux`, $0.05/MP) is Cova's cheapest "restyle this room" fallback when the user wants an image-variation of an exemplar. **InstantStyle and StyleAligned** don't cleanly port to DiT/flow-matching models; skip.

Section 5 — Aesthetic maximization strategy (elevated priority)

5.1 LoRA stack commit

Cova's aesthetic ceiling is set here. The CivitAI catalog is skewed toward generic luxury LoRAs rather than clean per-archetype adapters; **there is no canonical Japandi / Biophilic / Coastal FLUX LoRA of production quality on HuggingFace or CivitAI as of April 2026.** Cova must train its own.

Recommended stack per render (FLUX.1-dev on `fal-ai/flux-general`, $0.075/MP):

```
archetype_weights = softmax_top3_blend(target_6d, anchors, τ=0.3)
loras = [
  {path: cova_archetype_A, scale: archetype_weights[A]},        # 0.3–1.0
  {path: cova_archetype_B, scale: archetype_weights[B]},
  {path: cova_archetype_C, scale: archetype_weights[C]},
  {path: SedatAl/Interior-Flux-Lora,            scale: 0.3},     # grounding
  {path: mnml-ai/flux-arch-realism-lora,        scale: 0.5},     # polish
  {path: XLabs-AI realism (optional),           scale: 0.3},     # polish
]
ip_adapter = {image_url: exemplar_hero, scale: 0.5}
reference_image_url = exemplar_hero, strength: 0.45, start: 0%, end: 60%
```

Total effective LoRA weight ~1.8, fits fal-general's tolerance. The 0.7/0.3 archetype/polish pattern generalizes from CivitAI's well-documented Phlux+Amateur-Photo stacking (civitai.com/articles/6992).

5.2 Custom LoRA training — budget commit

The brief's "$150/LoRA on Replicate" is outdated: **Replicate Ostris flux-dev-lora-trainer runs $1–3 per default run at $0.001528/sec on H100; `fal-ai/flux-lora-fast-training` is flat $2 per run; `fal-ai/flux-2-trainer` is $0.008/step ≈ $8/1000-step run.**

Plan	Trainer	Per-archetype iterations	9-archetype total
Budget FLUX.1	fal flux-lora-fast-training	2	**$36**
Quality FLUX.1	fal flux-pro-trainer ($12/run)	2	**$216**
Budget FLUX.2	fal flux-2-trainer (1000 steps)	2	**$144**
Dual base (FLUX.1 + FLUX.2) hedged	mixed	1 each + iter	**$150–200**

Recommended: dual-base hedge at $150–200 total, lets Cova A/B on both generations before committing. **40–80 curated exemplar images per archetype** (denser than character LoRAs because interiors have more visual variance). fal's style-LoRA guide recommends 500–2000

steps, 1500 often best. **FLUX.2 LoRAs are not backward compatible with FLUX.1** (BFL HF discussion #6 on FLUX.2-dev confirms) — retrain from original data.

Exemplar library — 27 files (3 per archetype): hero wide-shot + material close-up + lighting beat. Unsplash Standard license is commercial-safe (avoid recognizable brand/art); **Unsplash+ explicitly excludes AI/ML use**; Pinterest individual pins are third-party copyrighted and cannot be ingested directly. Architectural Digest / Kinfolk / Dezeen / Studio McGee appear only as textual style invocations, never as training data. Each exemplar carries a JSON sidecar with archetype, source, license, role, 6D fingerprint vector, materials list, lighting descriptor, palette hex, keywords, and avoid list.

5.3 Prompt engineering — templates to ship

Universal editorial vocabulary that measurably lifts FLUX output toward magazine quality: `"architectural digest photography"`, `"kinfolk aesthetic"`, `"studio mcgee interior"`, `"dezeen minimalism"`, `"dwell magazine editorial"`, `"elle decor portfolio shot"`, `"the world of interiors photography"`.

Material specificity is what FLUX.1/2 reward most: `"white oak with visible open-pore grain, matte varnish"`, `"honed travertine, warm beige veining"`, `"ivory bouclé over linen"`, `"unlacquered brass with nascent patina"`, `"handwoven sisal rug"`, `"cerused oak"`, `"fluted cane"`, `"Venetian polished plaster"`, `"reeded walnut"`, `"limestone flags"`, `"raw concrete with oxide bloom"`, `"terracotta cotto tile"`, `"washed linen, natural crease"`.

Universal negative: `"warped perspective, floating objects, impossible geometry, fake symmetry, plastic textures, CGI sheen, oversaturated, HDR halos, melting edges, duplicated furniture, text overlays, watermark, people, cartoonish, overexposed highlights, muddy shadows, tiled repeating patterns, IKEA catalog, render artifacts"`.

Warm Japandi (60-word canonical template): `"Warm Japandi living room, kinfolk editorial: low-profile oak sofa with ivory bouclé cushions, shoji-paper sliding screens, tatami inlay, black ceramic vases, single ikebana branch, live-edge white oak coffee table, raw linen curtains, soft clerestory daylight, wabi-sabi, muted beige-charcoal-cream palette, matte black steel accents, 35mm f/5.6, layered depth, architectural digest."`

The other eight archetypes follow the same pattern (Biophilic Jungle, Coastal Bright, Dark Moody Glam, Heritage Traditional, Industrial Loft, Maximalist Eclectic, Pure Minimal White, Retro Mid-Century) with per-archetype lighting, material, and palette tokens laid out in the research bundle. Template strings live in `packages/director/prompts/archetypes/*.ts`, one file per archetype, gated by the forbidden-name linter (Section 7).

5.4 Post-processing — the polish chain

Generate at 1536×1024 (1.5 MP, $0.113 on flux-general), then `histogram_match(exemplar_hero)` → CLAHE with clip=2.5 tile=8 on LAB-L → per-archetype 3D LUT (`warm_japandi.cube`, `dark_moody_glam.cube` crushing blacks with teal-orange split-tone, etc.) → film grain σ 1–4 archetype-dependent → **`fal-ai/aura-sr` (AuraSR v2) 2× upscale (~$0.05)** → ship. AuraSR v2 outperforms Real-ESRGAN on interiors in fal's own benchmarks, tile-blended to suppress seams. **End-to-end render cost ≈ $0.16**, leaving ~$0.34 headroom under the cap for a second-pass refinement.

5.5 6D fingerprint → LoRA weight mapping

Baseline v1: softmax top-3 blend over archetype anchors with τ=0.3. Each archetype LoRA has a known 6D anchor (e.g., Heritage = `(0.80, 0.80, 0.30, 0.80, 0.95, 0.90)`, Pure Minimal White = `(0.20, 0.15, 0.15, 0.15, 0.20, 0.30)`). Per-axis fine control via `fal-ai/flux-general`'s `scale: {"layer_name": float}` dict for per-layer weighting when the research layer-axis mapping matures (`warmth` → final color blocks; `visual_complexity` → mid blocks; `organicness` → early-mid; `density` → late cross-attention; `traditionalism` → text cross-attention; `character` → early blocks).

Research support: **OSRM (ACL 2025, arXiv 2505.22934)** constrains LoRA subspaces so archetype updates stay orthogonal — if Cova trains archetype LoRAs with OSRM-style orthogonal init, linear interpolation becomes safe without concept-collapse. **Lee et al. arXiv 2507.11986** shows flow-matching latent fusion preserves style fidelity better than weight-merge for ⩾3 LoRAs. **Huang arXiv 2511.01355** gives a Content-Style Balance loss that rescues content fidelity at extreme style intensities — cap `archetype_scale ⩽ 1.1` and auto-insert the archviz realism polish LoRA at 0.25 when any slider exceeds 0.85. **LoRA-Composer (arXiv 2403.11627)** and **Multi-LoRA Composition (arXiv 2402.16843)** provide training-free alternatives worth benching.

Section 6 — Keep/remove for v1

V1 constraint (user-stated): only the render pipeline and refinement screen. The keep/remove UI proper is v1.1, but v1 must still *classify* what to preserve, because ghost-furniture gating and composite paste-back require it.

Auto-detection pipeline runs once at upload, before the user ever sees a screen:

1. **SAM 3 text-prompted pass** over `["ceiling fan", "framed photograph", "picture frame", "mirror", "fireplace", "crown molding", "exposed brick", "exposed beam", "wainscoting", "arched doorway", "bay window", "built-in bookshelf", "wall letters", "wall text"]` — returns candidate masks per concept with confidence.

SAM 3 documents that atomic concepts are reliable and compositional concepts (e.g., "family photo") degrade; use MLLM to first bbox-propose those, then SAM 3 refine from box.

2. **MediaPipe face detection inside each picture-frame bbox** — any face with confidence > 0.3 flags the frame as "family_photo → auto-keep". Fallback to RetinaFace on low-confidence.
3. **PaddleOCR over wall regions** — text ⩾4 characters at confidence ⩾0.5 flags the region as "wall_text → auto-keep" (the DAKSH-letters case).
4. **Architectural uniqueness** — OR-logic between SAM 3 open-vocab detection (step 1) and CLIP zero-shot as a cheap consensus; anything detected is lifted into the "must-preserve architectural mask" alongside walls/floor/ceiling/windows/doors.

The output is a structured `preservation_spec.json` with per-item masks, labels, and keep rationale, stored in Supabase and surfaced in the refinement screen as chips the user can toggle off ("I actually want the ceiling fan replaced"). No explicit tap-to-keep UI ships in v1.

Composite paste-back after redesign: LAB-L histogram match on the kept object to its target neighborhood (±25 clamp), 8px Gaussian-feathered alpha as the baseline, then conditional `cv2.seamlessClone`:

```python
def paste_back(bg_new, obj_crop, obj_mask, bbox, texturey):
    nbhd = extract_neighborhood(bg_new, bbox, pad=20)
    obj_lab = lab_L_histogram_match(obj_crop, nbhd, clamp=25)
    flag = cv2.MIXED_CLONE if texturey else cv2.NORMAL_CLONE
    try:
        out = cv2.seamlessClone(obj_lab, bg_new, dilate(obj_mask, 3), bbox_center(bb
        if seam_metric(out, bg_new, bbox) > 20: raise RuntimeError
        return out
    except Exception:
        alpha = feather(obj_mask, sigma=8)
        return alpha_blend(obj_lab, bg_new, alpha, bbox)
```

`NORMAL_CLONE` **for framed items** (family photos, art, wall text — preserves sharp edges). `MIXED_CLONE` **for contextual/textured objects** (ceiling fans against stucco — max-gradient blending preserves background texture). Neural compositing (IMPRINT, ObjectStitch) stays on the shelf as a v1.x rescue; the classical stack is deterministic, sub-second, and identity-preserving by construction.

Section 7 — Drift detection and quality gates

7.1 Metric stack (cheap-first)

Run sensors in this order, stop at the first failure:

1. **Window-bbox SSIM** (per-window IoU ⩾ 0.90, luma-only SSIM ⩾ 0.82 on bbox-tight crops with 10–15% padding).
2. **Whole-architecture SSIM** on the preservation mask (⩾ 0.92 pass, 0.85–0.92 soft-drift → MLLM arbitrate, <0.85 hard-drift → rescue).
3. **YOLO-World duplicate-furniture detection** over `["bed","dining table","kitchen sink","fireplace","toilet","bathtub","refrigerator","stove","ceiling fan","couch"]` at conf 0.35 / IoU 0.45 — any post-redesign count exceeding source count for a bed/fireplace/sink/toilet/bathtub/refrigerator/stove flags drift. Runs ~30–50ms on A10G.
4. **PaddleOCR watermark detection** with fuzzy edit-distance ⩽ 2 against `["collov", "reimagine", "decorilla", "modsy", "homedesigns", "©", "watermark"]` — any hit auto-re-rolls with "no logos, no watermarks, no text overlays" appended. ~80ms on A10G.
5. **CIEDE2000 ΔE** on wall regions — ΔE00 < 3 when a wall-repaint was requested means the edit didn't happen; ΔE00 > 25 paired with SSIM drop signals structural damage. ΔE00 ⩽ 1 is JND; 1–2 trained-eye-only; 2–3.5 commercial acceptability limit; ⩾5 clearly visible.
6. **DINOv2 cosine similarity** ⩾ 0.90 for "same scene, minor edit"; 0.80–0.90 "same room concept"; <0.75 concept drift.
7. **HPSv2.1 aesthetic score** — primary ranker when generating N=4 candidates; single forward pass ~60ms on A10G. v2.1 exceeds single-human agreement on HPD v2.
8. **AugCLIP (arXiv 2410.11374)** as text-conditioned oracle — gate behind cheaper checks because it needs an MLLM call per evaluation.
9. **MLLM judge (Haiku 4.5 vision)** — only when top-2 HPSv2 scores are within 0.5% or when sensors disagree. 5-axis rubric (style_match, architectural_fidelity, furniture_plausibility, lighting_coherence, composition) with 1–5 scoring, **pairwise comparison with randomized order** to neutralize position bias, and frozen rubric across iterations to prevent metric gaming.

7.2 Rescue economics

25% rescue rate × 90% rescue success × 2× cost of rescue = **~1.5–1.65× blended base cost**. At $0.16 primary, blended ≈ $0.25 — well inside the $0.50 cap. **Retry budget: 2 (initial + 1 rescue).** A

third attempt requires a structural path change (swap model class or downgrade scope to per-region refinement) and is logged with MLLM justification.

7.3 CoHP bounded iteration

Cap at 4 iterations (diminishing returns after 3 in MLLM-judge literature). Ship threshold: `min(score) ≥ 4 AND mean(score) ≥ 4.3`. After 4 iterations, soft-ship best-of-four and log `CoHP_softship` to telemetry. Judge prompt is frozen across iterations; only generator prompt evolves.

7.4 Forbidden-style-name enforcement

Five layers of defense against `"Japandi"` (alone), `"Bohemian"`, `"Coastal"` (alone), `"Mid-Century Modern"`, `"Minimalist"`, `"Scandinavian"`:

1. **ESLint custom rule** `cova/no-forbidden-styles` in `.ts/.tsx/.md/.json`, with an `allowRegex` for qualified forms (`Warm Japandi`, `Coastal Bright`).
2. **Husky + lint-staged pre-commit hook** scanning diffs.
3. **Runtime prompt sanitizer** wrapping `buildPrompt()` — throws `ForbiddenStyleError` if any forbidden token reaches the string.
4. **Zod schema** `.refine` on archetype/style/keyword fields in the DesignBrief.
5. **Log redactor** — Datadog/PostHog telemetry strips forbidden substrings before write.

Test coverage: every forbidden token standalone throws, every canonical archetype passes, compound strings ("Japandi Coastal Coastal") handled per spec.

Section 8 — Refinement screen (in-scope for v1)

8.1 UX commit

Region selection primary = tap-to-SAM (SAM 2 browser-hybrid: Modal A10G encodes once at upload, cached embedding in Supabase with 30-day TTL, ONNX Runtime Web decoder runs client-side at <50ms per tap). Single tap → coherent object mask; click-again adds positive seed; shift-click adds negative. **Brush as secondary** for free-form regions. **No lasso** — painful on touch devices. 44px minimum touch targets.

Prompt scope: region prompt operates within whole-scene context, pre-filled with `"{archetype_tokens}, {region_label}"` (Photoshop Generative Fill / MJ Remix pattern). User appends edits.

Variation count: 4 for whole-scene, 2 for region (cost + screen real-estate).

Undo/redo: 20-state stack per image; selection, prompt, and generation are discrete steps.

Comparison view: before/after swipe-handle (essential on mobile) + 2×2 grid for variations. Double-tap to zoom.

8.2 Backend

```
Client:  tap-to-SAM2 (decoder ONNX) → mask
Server:  dilate mask +8px, extract 1.1× bbox with padding
API:     fal-ai/flux-kontext-lora/inpaint @ $0.035/MP × 2 variants
Gate:    SSIM + DINOv2 (same drift stack)
Merge:   alpha-paste unchanged pixels from original outside dilated mask (2px
feather)
Return:  URLs to client
```

Preserving unchanged pixels is **not** automatic with diffusion fill — even outside the mask, re-encoding softens slightly. The alpha-paste step guarantees byte-identical unedited pixels except for a 2px feather boundary; CI test asserts hash equality.

Cheaper than FLUX.1 Pro Fill ($0.05/MP); still escalable to Pro Fill on second rescue. **Target latency < 8s per region edit**; cost ~$0.07/region × 2 variants.

Section 9 — End-to-end pipeline commit

9.1 Architecture

```
Upload (Next.js client)
  ↓
Modal A10G: image encode + preservation_spec detection
    SAM 3 open-vocab (architecture + keep-classes)
    MediaPipe face in frames
    PaddleOCR wall text
    Depth-Anything-V2 (cached for rescue)
    SAM 2 embeddings cached in Supabase (refinement)
  ↓
Empty-room (Bria primary / Canny+FLUX secondary / LaMa tertiary)
  ↓
Design Director (Sonnet 4.6 → Haiku 4.5 → 6D→archetype softmax)
  ↓
Redesign (NB-Pro primary / FLUX.2 Pro Edit fallback / FLUX-general + LoRA stack
when FLUX selected)
  ↓
```

```
Aesthetic post (CLAHE + LUT + grain + AuraSR v2)
  ↓
Keep composite (LAB-L match + seamlessClone NORMAL/MIXED)
  ↓
Drift gate (window SSIM → arch SSIM → YOLO dup → OCR → DINOv2 → HPSv2.1)
  ↓ fail → rescue (≤1 retry)
  ↓ pass
CoHP LLM-judge (4-iter cap, pairwise randomized)
  ↓
Return 4 candidates → refinement screen
  ↓
Region refinement loop (SAM 2 tap → flux-kontext-lora/inpaint → alpha preserve)
```

9.2 Per-stage spec (primary path)

Stage	Endpoint	Params	Cost	Latency	Fallback
Perception	`fal-ai/sam-3/image`	open-vocab prompts, $0.005/request × ~3 calls	$0.015	1–3s	SAM 2 + GroundingDINO self-host
Empty-room	`fal-ai/bria/eraser`	mask=SAM3 union +16px dilation	$0.04	5–15s	Canny+FLUX-general $0.08 / LaMa Modal ~$0.002
Director	Sonnet 4.6 + Haiku 4.5	existing; 6D→softmax top-3	~$0.02	2–4s	cached template
Redesign	`fal-ai/nano-banana-pro/edit`	1K, shell + 1–3 exemplars, high_thinking=false	$0.15	15–30s	FLUX.2 Pro Edit $0.03–0.06 / FLUX-general + LoRA stack $0.113
Post	AuraSR v2 + client LUT/grain	2× upscale	$0.05	2–5s	skip
Compose	Modal (CPU)	LAB-L + seamlessClone	~$0	<1s	feathered alpha
Drift gate	Modal A10G	SSIM + YOLO-World + PaddleOCR + DINOv2 + HPSv2.1	~$0.005	1–2s	—

Stage	Endpoint	Params	Cost	Latency	Fallback
Total primary			**~$0.28**	**~30–60s p50**	

9.3 Aesthetic stack spec

- Base: FLUX.1-dev on `fal-ai/flux-general` OR NB-Pro on `fal-ai/nano-banana-pro/edit` (A/B per bench results).
- LoRAs (FLUX path): 3 archetype softmax-blended (custom trained) + SedatAl/Interior-Flux-Lora @0.3 + mnml-ai/flux-arch-realism-lora @0.5 + XLabs realism @0.3.
- IP-Adapter: XLabs-AI/flux-ip-adapter-v2 @ scale 0.5 + reference-only strength 0.45 start 0% end 60%.
- Exemplars: 27-file curated library; Unsplash Standard + commissioned; JSON sidecars with 6D/materials/lighting/palette/keywords.
- Prompts: 50–80-word per-archetype templates; universal editorial prefix; universal negative.
- Post: CLAHE (clip 2.5, tile 8) → per-archetype .cube LUT → film grain σ1–4 → AuraSR 2×.

9.4 Cost and latency budget

Path	$/render	Est. p50	Under $0.50?	Under 60s?
Primary (NB-Pro)	$0.28	30–50s	✅	✅ likely; bench to confirm
FLUX path (flux-general + LoRA stack)	$0.16	25–40s	✅	✅
Rescue (SAM 3 envelope + FLUX.1 Pro Fill)	$0.20	35–50s	✅	✅
Blended with 25% rescue at 1.6×	$0.32–0.45	35–55s	✅	✅ p50; p95 tight
Refinement per region	$0.07–0.10	5–8s	✅	✅

9.5 Phase 0 bench plan — what to actually run

8 rooms from the 20-room frozen bench (stratified per §3.4). Candidates:

- C1 NB-Pro alone (shell only)
- C2 NB-Pro + shell + 1 exemplar
- C3 NB-Pro + shell + exemplar + high_thinking=true
- C4 FLUX.2 Pro Edit with `@image1`/`@image2`
- C5 FLUX-general + Cova archetype LoRA stack + IP-Adapter + reference-only (once Cova LoRAs exist; until then, SedatAl baseline)
- C6 Kontext Max + IP-Adapter (control)
- C7 SAM 3 envelope mask + FLUX.1 Pro Fill
- C8 FLUX.1 + StableDesign-ported dual-CN (seg + depth) + binary-window-inpaint (the Lavreniuk port)

Measured against §3.4 architecture thresholds + §5 aesthetic metrics (palette coherence Bhattacharyya ⩽ 0.35; style fidelity DINOv2 ⩾ 0.72 vs exemplar; decor count ⩾12 via YOLO-World; HPSv2.1 ⩾ 5.8/10). **Pass = architecture AND window IoU AND latency AND cost pass on ⩾18/20 rooms, AND aesthetic metrics pass on ⩾16/20. Fail on any architectural room disqualifies.**

If C1/C2/C3 wins outright: commit NB-Pro primary. **If C4 is within 5% on aesthetics and cheaper + faster:** commit FLUX.2 Pro Edit as primary, NB-Pro as quality-tier rescue. **If C5 wins only after custom LoRAs:** Phase 4 (LoRA training) gates Phase 5 primary selection; ship FLUX.2 Pro Edit as interim primary. **If no candidate clears ⩾7/8 architectural pass:** Phase 5 does not ship; reopen the question.

9.6 Honest confidence statement

What I'm ~95% confident about:

- Empty-room stage primary = Bria Eraser (prompt-free, $0.04 flat, addresses Rounds 2.5/4 directly).
- SAM 3 text-prompts resolve Round 9's synthetic-canvas failure.
- Per-window IoU ⩾ 0.90 gating + auto-rescue catches architectural drift before ship.
- Forbidden-style enforcement at 5 layers prevents name leakage.
- Keep-composite classical stack (LAB-L + seamlessClone + feathered fallback) is deterministic and identity-preserving.
- Cost math fits $0.50 cap with ~2× headroom on all paths.
- $150–200 total budget for 9 archetype LoRAs is correct (not the $1350 in the brief).

What I'm ~70% confident about:

- NB-Pro beats FLUX.2 Pro Edit on architectural preservation with shell + exemplar multi-ref. The reasoning story is compelling but empirical gap exists. **This is the single most important bench question.**
- NB-Pro p50 on fal queue clears 60s. fal does not publish SLA for Pro models; if bench p95 > 60s demote to Flash (NB2).
- Cova's custom FLUX LoRAs will beat off-the-shelf generalist LoRAs on the 6D fingerprint sliders. Almost certainly true at the extremes; middle-of-slider is untested.
- MLLM judge pairwise (Haiku 4.5) is stable enough to be the CoHP tiebreaker. Known narcissistic bias (~25% for Claude on Claude outputs) requires we not use Claude-generated captions as judge input.

What still needs Phase 0 validation:

- Whether FLUX-general's single-ControlNet limit + Union Pro 2.0 internal multi-mode + IP-Adapter slot stack is sufficient, or whether Cova must self-host a true multi-CN pipeline on Modal A10G.
- Whether SAM 3 self-hosted license permits commercial use or only the fal endpoint does — affects Modal deployment plan.
- Whether Depth-Anything-V2 + Canny-from-depth secondary fallback achieves its SSIM ≥ 0.985 threshold on the "geometry difficult" bench rooms (long perspective corridor, wall-corner-in-frame).
- StableDesign port (C8) — no one has published dual-CN + binary-window inpaint on FLUX; this is original work and might or might not clear the bench.
- Exemplar library licensing — Unsplash Standard is fine for exemplars; if any of the 27 images enter LoRA training data, double-check the "training a similar or competing service" prohibition in the Unsplash license.

What I refuse to claim 100% confidence about, honestly: that any static pipeline description will produce ship-quality on the first Claude Code pass across all 20 bench rooms. The correct production stance is **the Phase 0 bench IS the 100% confidence gate**, not this document. Every prior recommendation in this session failed because it was committed without empirical gating; this one commits only after §9.5 passes.

Section 10 — Phase plan

Each phase lists scope, deliverables, acceptance, budget (API $ + days), dependencies, parallelization, and Claude Code readiness.

Phase 0 — Codebase audit. Scope: the six audit tracks in §1. Deliverables: six `docs/audit/*.md` files; dead code removal PRs; failing RLS negative tests converted to green; LoRA kept/dropped list. Acceptance: all six deliverables landed, merged, reviewed. Budget: $0 API, **3–5 days**. Dependencies: none. Parallelization: all six tracks in parallel. Claude Code readiness: **immediate** — each track is a bounded audit task.

Phase 1 — Perception service (Modal A10G). Scope: SAM 3 wrapper + SAM 2 browser-hybrid encoder + Depth-Anything-V2 caching + PaddleOCR + MediaPipe face. Deliverables: `modal/perception.py` with `/detect_preservation_spec`, `/encode_sam2_for_cache`, `/depth_anything_v2` endpoints; Supabase `sam_embeddings` table + RLS; embedding cache writer. Acceptance: bench image → preservation_spec.json in ⩽ 3s p95; SAM 2 embedding cache <2MB/image. Budget: ~$0.02/call × bench, **4 days**. Dependencies: Phase 0 RLS audit. Parallelization: SAM 3 + SAM 2 + depth in parallel subtasks.

Phase 2 — Empty-room service. Scope: Bria Eraser orchestration + Canny secondary + LaMa tertiary + ghost gate. Deliverables: `modal/empty_room.py`; ghost-detection rerun with SAM 3; SSIM/VP-shift metric functions; fallback routing; bench runner for 8 rooms. Acceptance: ⩾7/8 rooms pass §3.4 thresholds. Budget: $0.05–0.15/call × ~200 bench invocations = ~$30, **5 days**. Dependencies: Phase 1.

Phase 3 — Design Director rewrite. Scope: STYLE_DNA_BLOCKS → softmax-blend migration; 6D→archetype anchors; TS↔Python parity fold; forbidden-name linter + Zod refines; preservation clause vocabulary (borrow Decor8's whitelist). Deliverables: `packages/director/` single source of truth (choose TS or Python, fold the other); unit tests for 100 briefs → golden prompts; lint+Zod+log redactor stack; 9 archetype template files. Acceptance: 100/100 snapshot prompts stable; forbidden tokens never reach prompt; brief JSON passes Zod. Budget: ~$5 Sonnet/Haiku eval, **6 days**. Dependencies: Phase 0 director audit.

Phase 4 — Aesthetic stack. Scope: exemplar library curation (27 files + sidecars); custom LoRA training (9 archetypes × 2 iterations via `fal-ai/flux-lora-fast-training`); IP-Adapter integration on `fal-ai/flux-general`; LUT authoring per archetype; prompt templates. Deliverables: `exemplars/` library with JSON sidecars; `loras/cova_archetype_*.safetensors` on fal; LUT .cube files; archetype prompt template files; A/B eval on 8 bench rooms. Acceptance: LoRA-stacked render beats baseline on HPSv2.1 by ⩾0.3 per archetype. Budget: **~$200 training** + $50 eval API + image licensing if commissioned, **7–10 days**. Dependencies: Phase 0 LoRA audit (keep/dropped), Phase 3 templates. Parallelization: exemplar curation + LoRA training in parallel.

Phase 5 — Redesign service. Scope: NB-Pro primary; FLUX.2 Pro Edit fallback; FLUX-general + LoRA rescue; SAM 3 envelope mask + FLUX.1 Pro Fill as structural rescue. Deliverables: `modal/redesign.py` orchestrator; fallback router with metric-gated escalation; prompt compiler; reference image passing. Acceptance: **§9.5 bench pass**. Budget: $0.16–0.30/render × ~300 bench = ~$90, **7 days**. Dependencies: Phases 1, 3, 4.

Phase 7 — CoHP bounded iteration. Scope: 4-iter cap; frozen-rubric Haiku 4.5 pairwise judge; soft-ship on cap; telemetry. Deliverables: `modal/cohp.py` with pairwise randomized judge and position-bias neutralization. Acceptance: 4-iter cap enforced; soft-ship logs. Budget: ~$0.03 Haiku/iter × bench = ~$15, **4 days**. Dependencies: Phase 6.

Phase 8 — Refinement screen UI + backend. Scope: Next.js tap-to-SAM UI with ONNX Runtime Web decoder; brush secondary; undo/redo 20-state stack; swipe + 2×2 grid comparison; `fal-ai/flux-kontext-lora/inpaint` backend with alpha-preserve; 2-variant generation. Deliverables: `apps/web/app/refine/` route; `modal/refine.py`; CI hash-equality test; SAM embedding TTL job. Acceptance: region edit p50 ⩽ 8s; unchanged pixels hash-equal. Budget: ~$0.07/region × test = ~$30, **8–10 days**. Dependencies: Phases 1, 6. Parallelization: UI and backend in parallel.

Phase 9 — Integration wiring. Scope: web → Modal → fal orchestration; Supabase Edge Function nightly aggregator extension; retry budget enforcement. Deliverables: `services/pipeline.ts` orchestrator; E2E integration test harness on 8 rooms. Acceptance: end-to-end p50 ⩽ 60s; cost ⩽ $0.50; all fallbacks exercised. Budget: ~$50 integration, **4 days**. Dependencies: Phases 2, 5, 6, 7, 8.

Phase 10 — Test suite extension. Scope: ~40 new test cases enumerated in §1 audit. Deliverables: tests in `__tests__/pipeline/`; GitHub Actions config adjustments; nightly golden-room regression. Acceptance: all new tests green; 715-test baseline regressions = 0. Budget: $0, **3 days**. Dependencies: Phases 2, 5, 6.

Phase 11 — Quality benchmark vs Collov. Scope: 20-room pairwise evaluation: Cova output vs Collov output on identical input rooms, MLLM-judge + HPSv2.1 + human review. Deliverables: `docs/benchmark/cova_vs_collov.md` with pairwise win-rate per archetype. Acceptance: Cova ⩾ 55% win rate on aesthetic; ⩾ 80% win rate on architectural preservation (structural fidelity is Cova's stated wedge). Budget: $0.30/render × 40 renders = $12, plus $5 Collov subscription for comparison renders, **3 days**. Dependencies: Phase 9.

Total timeline: 12 weeks if sequential, ~7–8 weeks with aggressive parallelization (Phase 0 tracks in parallel; Phases 4 exemplar + LoRA in parallel; Phases 8 UI + backend in parallel; Phase 11 overlaps with Phase 10). **Total API spend through Phase 11: ~$450–550**, dominated by LoRA training ($200) and redesign bench iterations ($90 + $50).

Section 11 — Risk register (top 10)

1. **NB-Pro p50 exceeds 60s on fal queue.** Probability 30%. Impact high (demotes primary). Early warning: Phase 5 bench p95 > 75s. Mitigation: Nano Banana 2 Flash ($0.08, 5–12s) as drop-in replacement; architecture unchanged.
2. **Custom LoRA training does not beat off-the-shelf on 6D middle-of-slider cases.** Probability 25%. Impact medium (falls back to prompt-only). Early: A/B HPSv2.1 delta < 0.2. Mitigation: OSRM orthogonal init on retrain; increase exemplar count to 80/archetype; latent-fusion composition per arXiv 2507.11986.
3. **SAM 3 self-host license blocks commercial use; only fal endpoint permitted.** Probability 20%. Impact low (cost rises slightly since SAM 3 runs per-render rather than per-session). Early: legal review in Phase 1. Mitigation: GroundingDINO + SAM 2 self-host fallback.
4. **Bria Eraser quality fails on heavy-clutter rooms.** Probability 20%. Impact medium (escalates to Canny+FLUX rescue more often, blended cost rises). Early: Phase 2 bench fails ⩾2/8 heavy-clutter rooms. Mitigation: promote Canny+FLUX secondary to primary for heavy-clutter auto-detected cases.
5. **FLUX-general single-ControlNet limit blocks StableDesign port quality.** Probability 30%. Impact medium (rescue path weakens). Early: Phase 5 bench C8 fails. Mitigation: self-hosted ComfyUI-style multi-CN on Modal A10G; adds ~3 days to Phase 5.
6. **SAM 2 browser-hybrid decoder latency on low-end phones > 200ms.** Probability 25%. Impact medium (refinement UX degrades). Early: Phase 8 field test on iPhone 12 / Pixel 5. Mitigation: server-side fallback for devices without WebGPU; ONNX Runtime Web quantized decoder.
7. **MLLM judge (Haiku 4.5) exhibits archetype bias, preferring some archetypes over others.** Probability 40%. Impact medium (CoHP ship rate drifts per archetype). Early: Phase 7 ship-rate delta > 15% across archetypes. Mitigation: per-archetype rubric calibration; pairwise instead of scoring; swap to GPT-4o vision for suspected-biased archetypes.
8. **Exemplar library licensing challenge post-launch.** Probability 15%. Impact high (library rebuild). Early: pre-launch legal review. Mitigation: commission original photography for hero exemplars; Unsplash Standard only for material close-ups with attribution record; never use Unsplash+ images.
9. **Duplicated TS + Python Director in §1 cannot be folded cleanly.** Probability 20%. Impact medium (Phase 3 delay). Early: TS↔Python parity diff shows >10% fields drift. Mitigation: pick Python on Modal as source of truth (server-side only); wrap with typed TS SDK generated from JSON schema.

10. **Phase 0 LoRA audit finds every existing LoRA fails evaluation.** Probability 30%. Impact low–medium (reinforces Phase 4 custom training requirement). Early: Phase 0 eval round. Mitigation: Phase 4 is already budgeted; Phase 5 primary commits to NB-Pro which doesn't depend on FLUX LoRAs.

Section 12 — Decision checkpoint summary

One-line recommended pipeline. `SAM 3` perception + `fal-ai/bria/eraser` empty-shell → Design Director 6D→archetype-softmax brief → **`fal-ai/nano-banana-pro/edit`** redesign with shell + 1–3 exemplar multi-ref (fallback `fal-ai/flux-2-pro/edit`; rescue SAM 3 envelope + `fal-ai/flux-pro/v1/fill`) → LAB-L + seamlessClone composite → SSIM/YOLO-World/PaddleOCR/DINOv2/HPSv2.1 drift gate with ≤1 rescue retry → 4-iter CoHP Haiku 4.5 pairwise judge → 4 candidates returned → SAM 2 tap-to-region refinement via `fal-ai/flux-kontext-lora/inpaint` with alpha-preserve.

Go/no-go criteria for Phase 5 primary commit. The Phase 0 bench of 8 stratified rooms must meet: SSIM-unmasked ⩾ 0.985, zero ghost-furniture > 0.5%, VP shift < 2%, per-window IoU ⩾ 0.90, p50 ⩽ 60s, cost ⩽ $0.50, HPSv2.1 ⩾ 5.8. Pass on ⩾7/8 rooms commits NB-Pro primary; 5–6/8 triggers FLUX.2 Pro Edit primary; <5/8 reopens Section 4.

What Phase 0 must produce to proceed. Six audit deliverables in §1 (LoRAs kept/dropped; Director parity + STYLE_DNA migration; Modal dead code removed; RLS negative tests green; new test plan; pgvector EXPLAIN). Without any one of these, Phase 5 must not ship.

Single biggest risk + mitigation. NB-Pro primary p50 exceeds 60s on fal queues (30% probability, high impact). Mitigation: pre-committed Nano Banana 2 Flash fallback with identical API shape; branch in orchestrator keyed on measured p50 from last 100 renders.

Honest confidence assessment.

- **95% sure:** empty-room primary (Bria), SAM 3 text-prompt resolves R9, keep-composite classical stack, forbidden-name enforcement, cost math fits cap, ~$200 total LoRA budget.
- **70% sure:** NB-Pro beats FLUX.2 Pro Edit on preservation with multi-ref; Cova custom LoRAs beat generalists on 6D sliders; Haiku 4.5 pairwise judge is stable enough as CoHP tiebreaker.
- **Still needs validation (Phase 0 bench):** NB-Pro p50 on fal queue; StableDesign FLUX port (C8); secondary Canny+FLUX on difficult geometry; SAM 3 self-host licensing; exemplar library license audit.

The commit is: this pipeline, gated by the Phase 0 bench, with fallbacks pre-wired. Every Round-1-through-11 failure mode has an explicit mitigation mapped in §4.1. The 100%

confidence the user asked for does not belong in this document; it belongs in the bench pass report that Phase 5 emits before shipping. Everything in this document is what Claude Code needs to begin Phase 0 immediately.